Course: ITAI-1371 – Introduction to Machine Learning

Lab: Module 07 Lab – Better Model Evaluation

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1. Lab Objectives & Overview

The core objective of this lab was to transition beyond raw overall accuracy and develop a rigorous understanding of multidimensional model evaluation. Using a Logistic Regression baseline trained on the Titanic survival dataset (Age, Pclass, Sex, and Fare), our team examined how classification models actually distribute errors between False Positives (Type I) and False Negatives (Type II). By implementing a confusion matrix, calculating precision, recall, and F1-score, and running 5-fold cross-validation, we analyzed the stability and practical limitations of our model.

2. Technical Implementation & Results Summary

- Data Preparation & Baseline:
- Cleaned missing numeric values using median imputation for Age.
- Encoded categorical features (Sex: Male = 0, Female = 1).
- Executed an 80/20 train-test split (test_size=0.2, random_state=42).
- Trained a LogisticRegression(max_iter=1000) classifier.

Confusion Matrix Breakdown (Test Set = 179 samples):

- True Negatives (TN): 90 (Did not survive correctly identified)
- False Positives (FP / Type I): 15 (Predicted survived, but actually died)
- False Negatives (FN / Type II): 20 (Predicted died, but actually survived)
- True Positives (TP): 54 (Survived correctly identified)

Classification Metrics:

- Overall Accuracy: 80% (144 / 179)
- Class 0 (Did Not Survive): Precision = 0.82 | Recall = 0.86 | F1-Score = 0.84
- Class 1 (Survived): Precision = 0.78 | Recall = 0.73 | F1-Score = 0.76
- Macro / Weighted Average: F1 = 0.80

5-Fold Cross-Validation (cv=5):

- Fold Accuracies: 78.77%, 78.09%, 78.65%, 77.53%, 80.34%
- Mean CV Score: 78.68%
- Standard Deviation: 0.0094 (±0.94%)

3. Critical Reflections & Key Takeaways

Accuracy Is Insufficient in Isolation

Our single split test accuracy showed an even 80.0%. While that headline number looks solid on paper, isolating accuracy hid the underlying disparity between classes. The model demonstrated a significantly higher recall for non-survivors (0.86) than for survivors (0.73). In critical applications, a 27% False Negative rate on the target class would represent an unacceptable blind spot that an 80% aggregate accuracy completely obscures.

The Precision-Recall Trade-Off in Real-World Contexts

Analyzing the mathematical formulas alongside practical scenarios clarified how business goals dictate metric priority:

- High-Precision Priority: In spam filtering or automated moderation, False Positives carry severe disruption costs (e.g., dropping critical client communications). A model must maintain high precision so that any positive assertion is trustworthy.

- High-Recall Priority: In clinical screening or high-stakes anomaly detection, False Negatives can be fatal. Catching every actual positive instance takes precedence, as follow-up confirmatory tests can mitigate False Positives.

- The F1-Score proved essential for our Titanic model because survival was non-trivial to predict, and neither precision nor recall alone reflected overall decision quality.

Robustness Through Cross-Validation

Our single train-test split yielded an 80.00% accuracy, whereas our 5-fold cross-validation averaged 78.68%. This ~1.3% variance highlighted how a single split can inadvertently capture a slightly favorable test partition. The low standard deviation (0.0094) across the 5 folds confirmed that the logistic regression model maintains stable generalization and isn't suffering from extreme variance or overfitting across data subsets.

4. Challenges & Troubleshooting

- Preprocessing Decisions: Median imputation for Age was effective to avoid skew from outliers, but it inevitably softened variance within passenger demographics.

- Metric Interpretation: Reconciling the 15 Type I errors versus 20 Type II errors required careful examination of decision thresholds. Because default logistic regression relies on a 0.5 decision threshold, adjusting that threshold would allow us to artificially steer the balance between precision and recall depending on the specific cost function.